AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 26 · STUDENT LESSON

Image and Voice Pipelines

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

Preserve the source. Trace the transformation. Bound the claim.

Locate supported speech/image pipeline mismatches without inventing unavailable evidence.

Keep one habit: Representation limit → visible image trace → unresolved versus authoritative cases → four-stage audit.

MAIN LESSON · SECTION 1

At which stage did the information change?

> Keep source evidence, intermediate text and final claims separate.

Your learning goals

I can explain the **mechanism** and terminology of speech **pipeline**s and multimodal **evidence**.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Recall representation versus factual evidence

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

- Why does an exact text round trip not prove that the represented statement is factually correct?

Your end goal

By the end, I can trace where speech information changes and resolve a declared-authority conflict.

▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Text as Pieces. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Recommendation Scores. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Repair Bias in Data. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 26: Image and Voice **Pipeline**s

Perception systems transform **signal**s through multiple fallible stages.



THINKER MISSION

Trace speech-to-command and image-to-label **pipeline**s.

Thinker name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

This week’s end goal

MAIN LESSON · SECTION 2

Trace a multimodal pipeline using supplied evidence

> These are classroom traces, not live image or speech-service runs.

The modality and error-propagation problem

A multimodal **pipeline** may transcribe speech, inspect an image and generate a response. Errors at one stage can propagate even when later stages run correctly.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

▼ **Earlier wording, alternatives and picture transcripts**

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



MYSTERY

The learners must use image and voice **pipeline**s to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

MAIN LESSON · SECTION 3

Keep image/audio source, extracted text and the task answer distinct

> Locate only the mismatch the supplied evidence supports.

The source/interpretation/final-claim record

Track source **modality**, segment/time references, **transcription**, interpretation and final claims separately. Compare conflicting **evidence** without assuming one **modality** always wins.

Ask for clarification when the task needs unresolved information.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

The actual supplied reference image



Trace the simulated image-reading error

OCR means extracting text from an image. The supplied reference image reads SHELF 14; our simulated OCR output is SHELF 40. Because the visible source is supplied, this case supports a specific extraction mismatch. A correct lookup of shelf40 afterward would still be based on the wrong extracted number. This is a constructed trace, not an actual OCR evaluation or a measured accuracy score.

Explain a pictured stage

- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Earlier wording, alternatives and picture transcripts

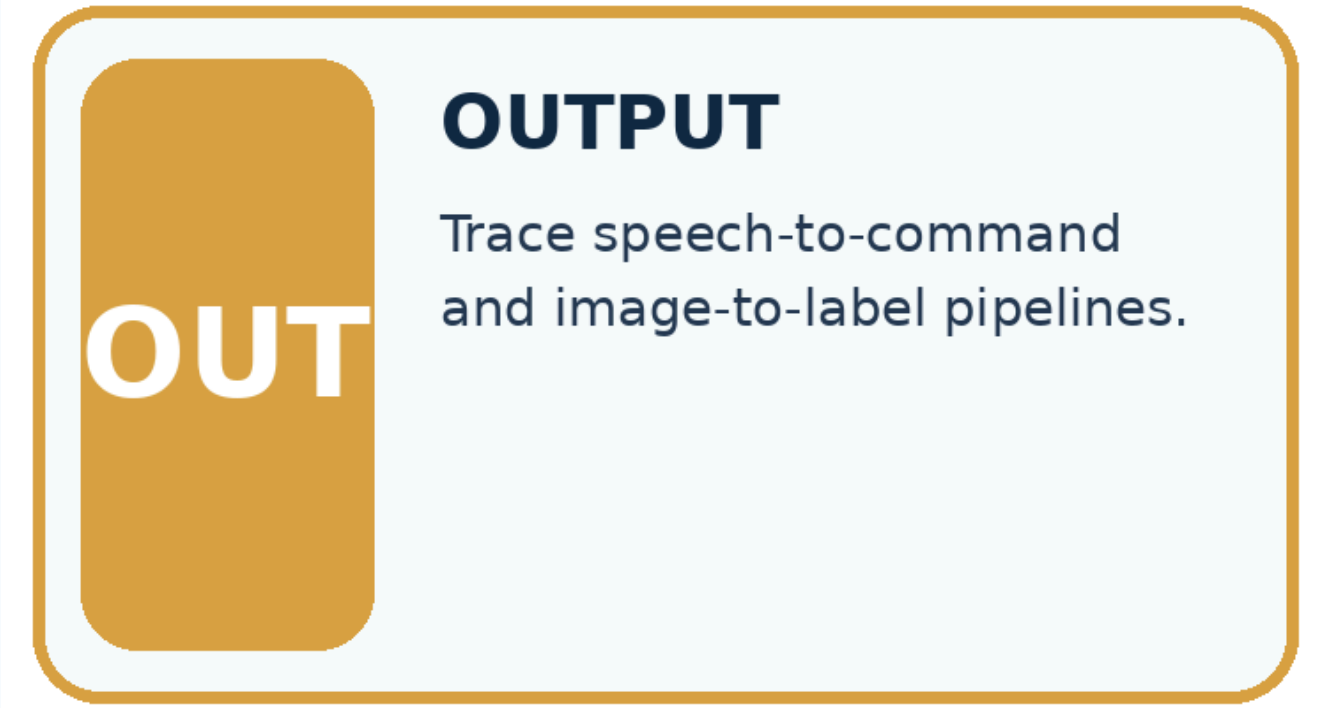
VISUAL **MECHANISM**

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.


IN
INPUT
Trace speech-to-command and image-to-label pipelines.
↓
↻
PROCESS
Perception systems transform signals through multiple fallible stages.

↓



Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

Speech pipelines and multimodal evidence — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Audio transcript reads fifteen boxes; a clear written inventory says fifty boxes. The original audio is unavailable and the assistant reports fifteen as certain.	Track source modality, segment/time references, transcription, interpretation and final claims separately. Compare conflicting evidence without assuming one modality always wins. Ask for clarification when the task needs unresolved information.	There is an unresolved quantity conflict and no audio replay. Preserve both sources and ask the responsible person; certainty is unsupported.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

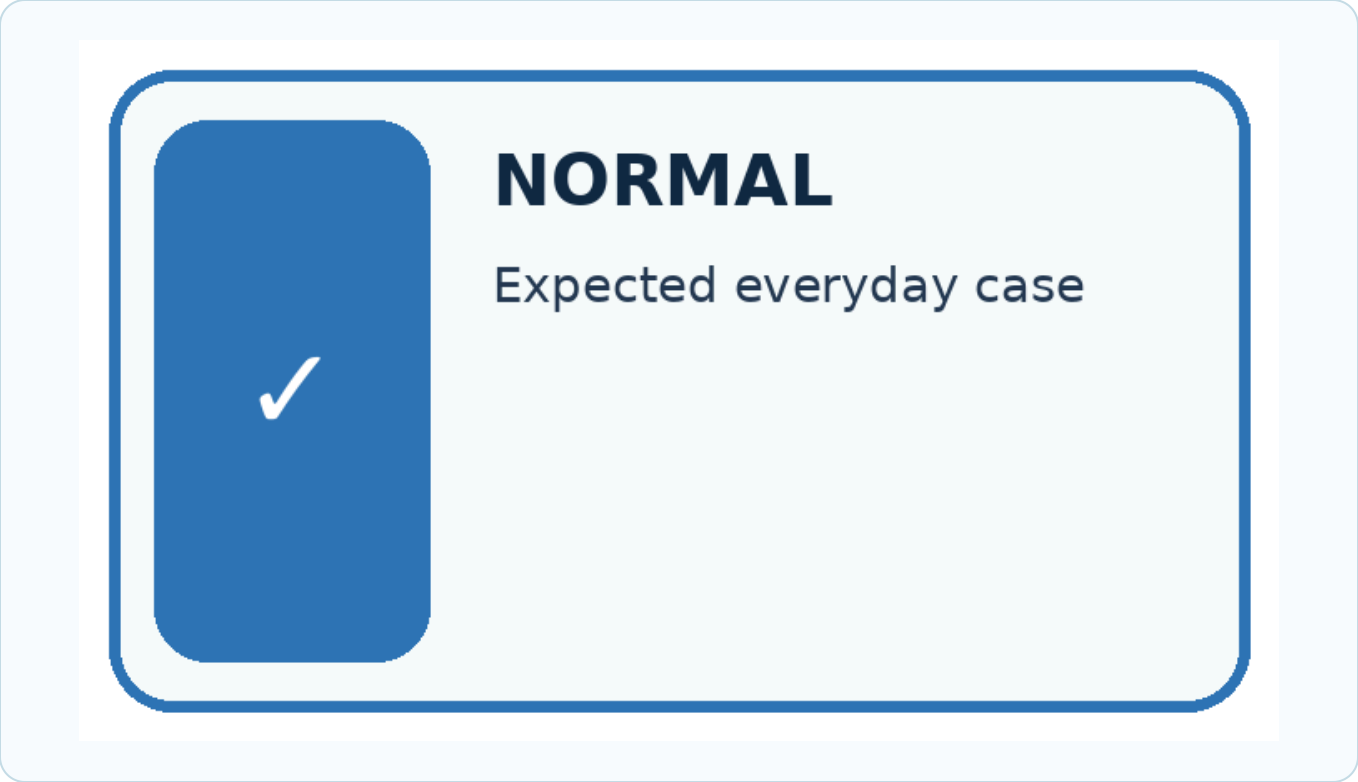
Connect the picture and the explanation

MAIN LESSON · SECTION 4

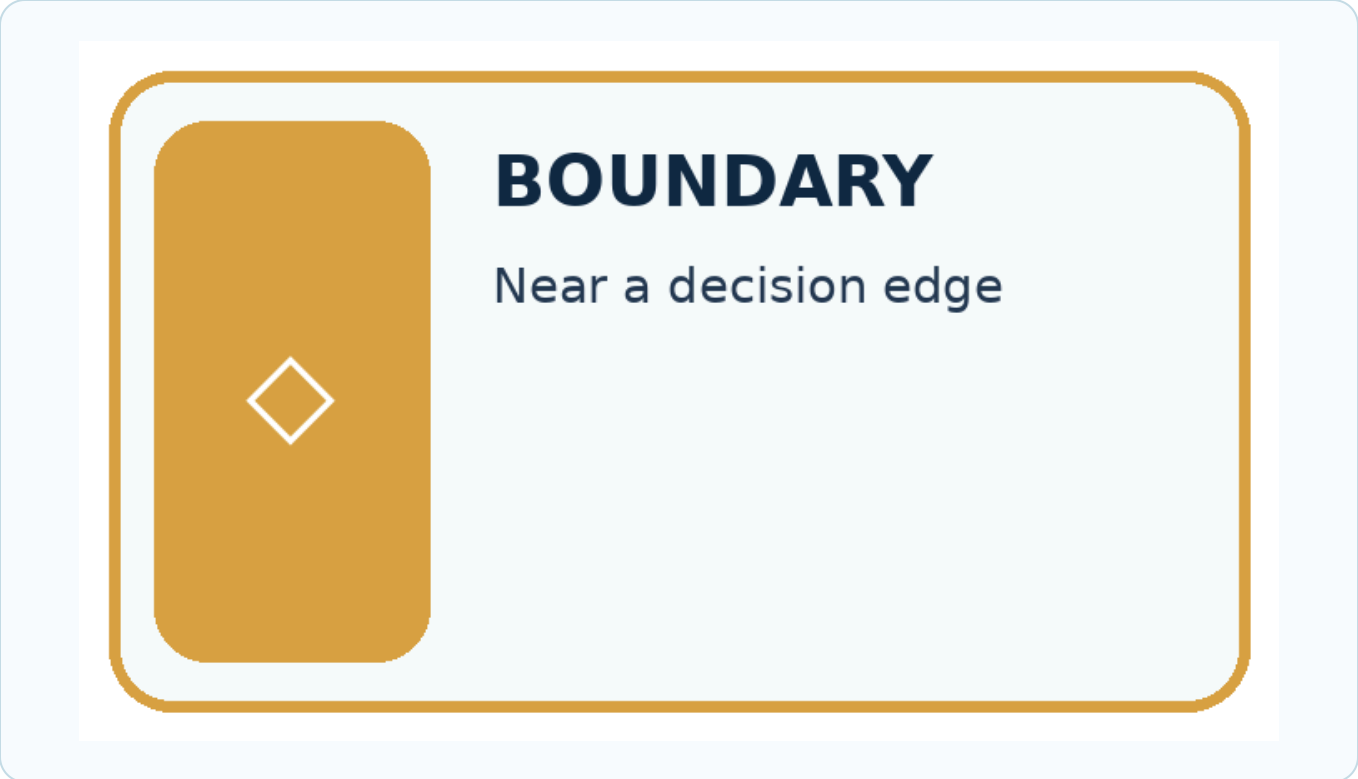
Compare replayable evidence with an unresolved source conflict

> No modality wins automatically; use the declared task and authority.

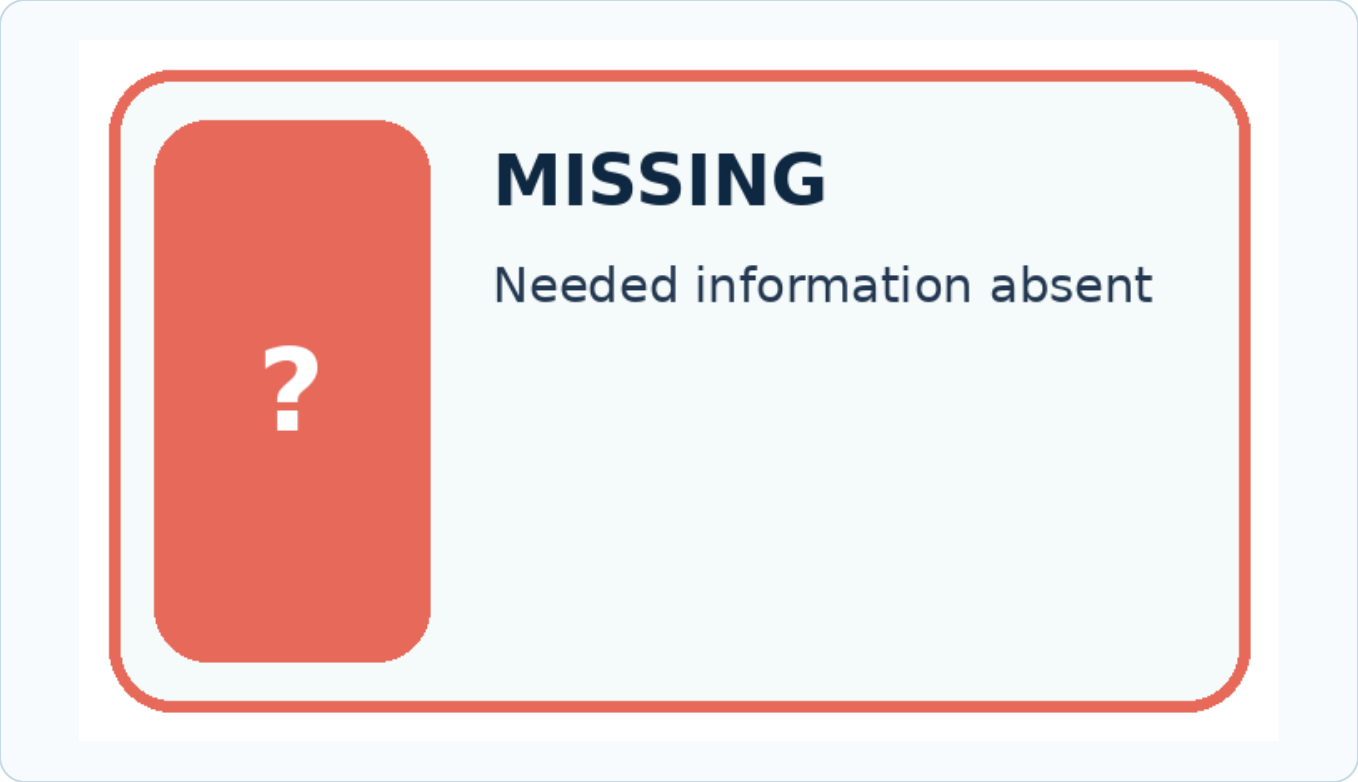
Four pictures with a purpose



Normal case: predict the result and deciding evidence.



Boundary case: identify the exact limit.



Missing information: name what must be clarified.



Transfer case: explain which procedure still applies.

Choose a revealing case

- Which case is most likely to reveal a hidden assumption? Why?

> **Write or draw your reasoning and evidence.**

Predict the supplied changed case

- Change: Remove access to the audio, while keeping the authoritative inventory.

▼ Check after predicting

Prediction: Shelf 14 remains supported for the lookup, but **transcription** accuracy cannot be replay-checked.

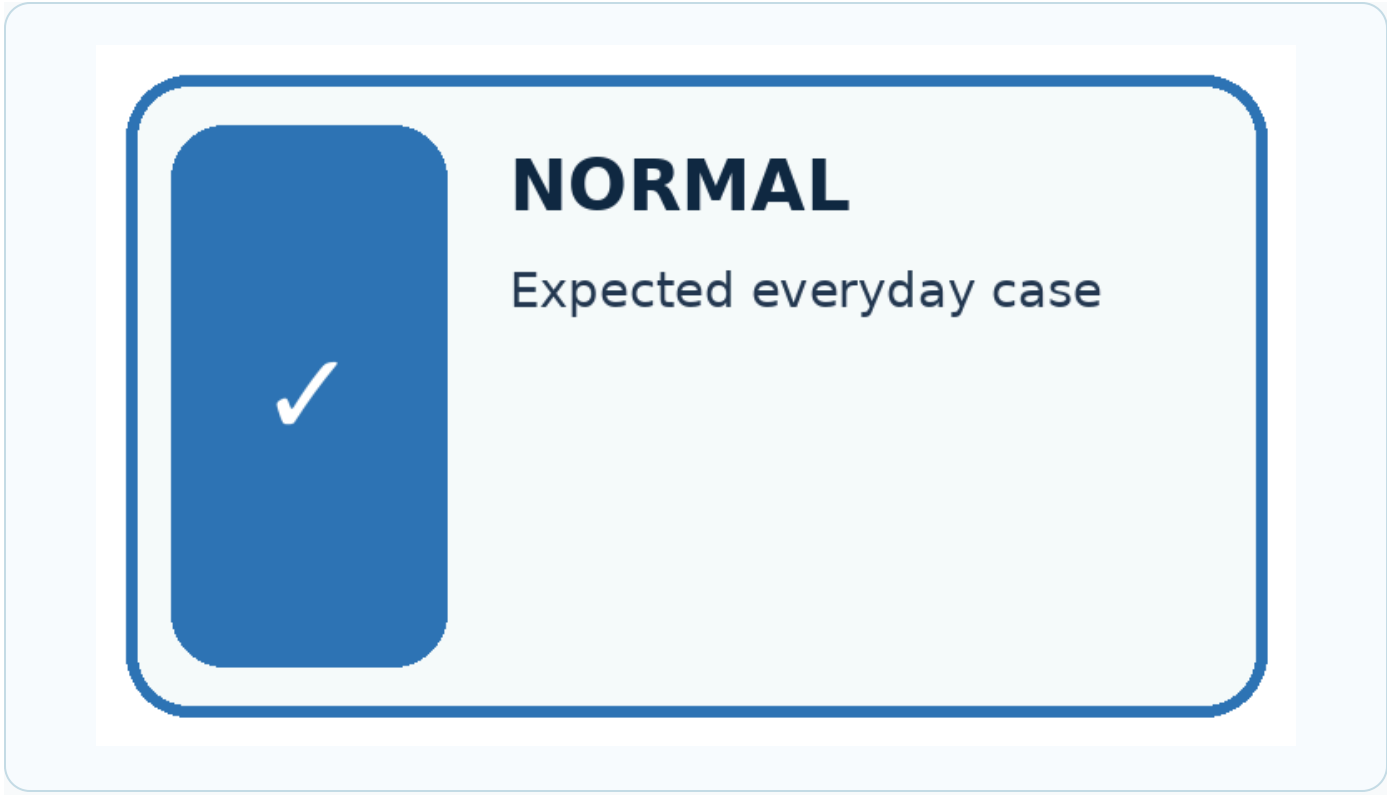
Reason: The inventory resolves the shelf task; it does not recreate unavailable audio.

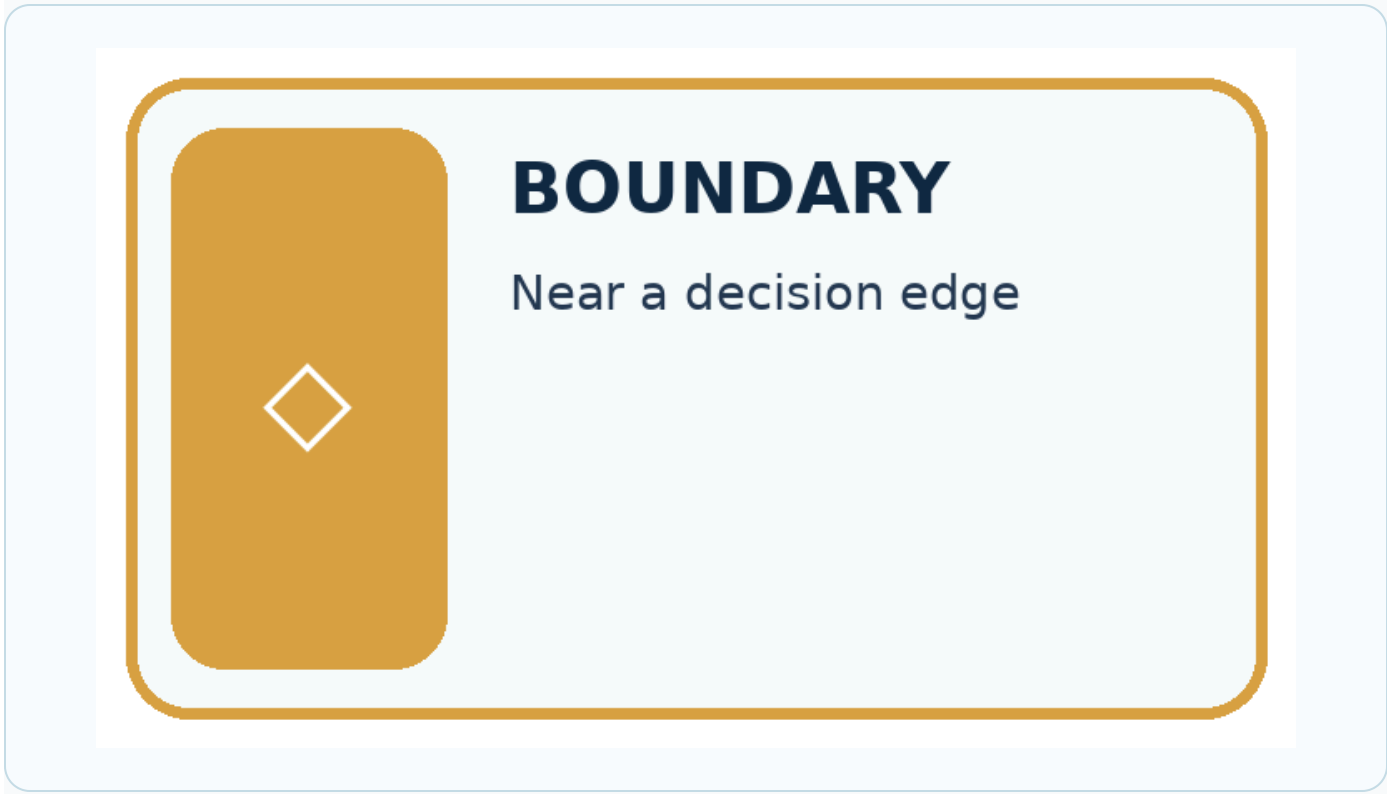
▼ Earlier wording, alternatives and picture transcripts

CASE GALLERY

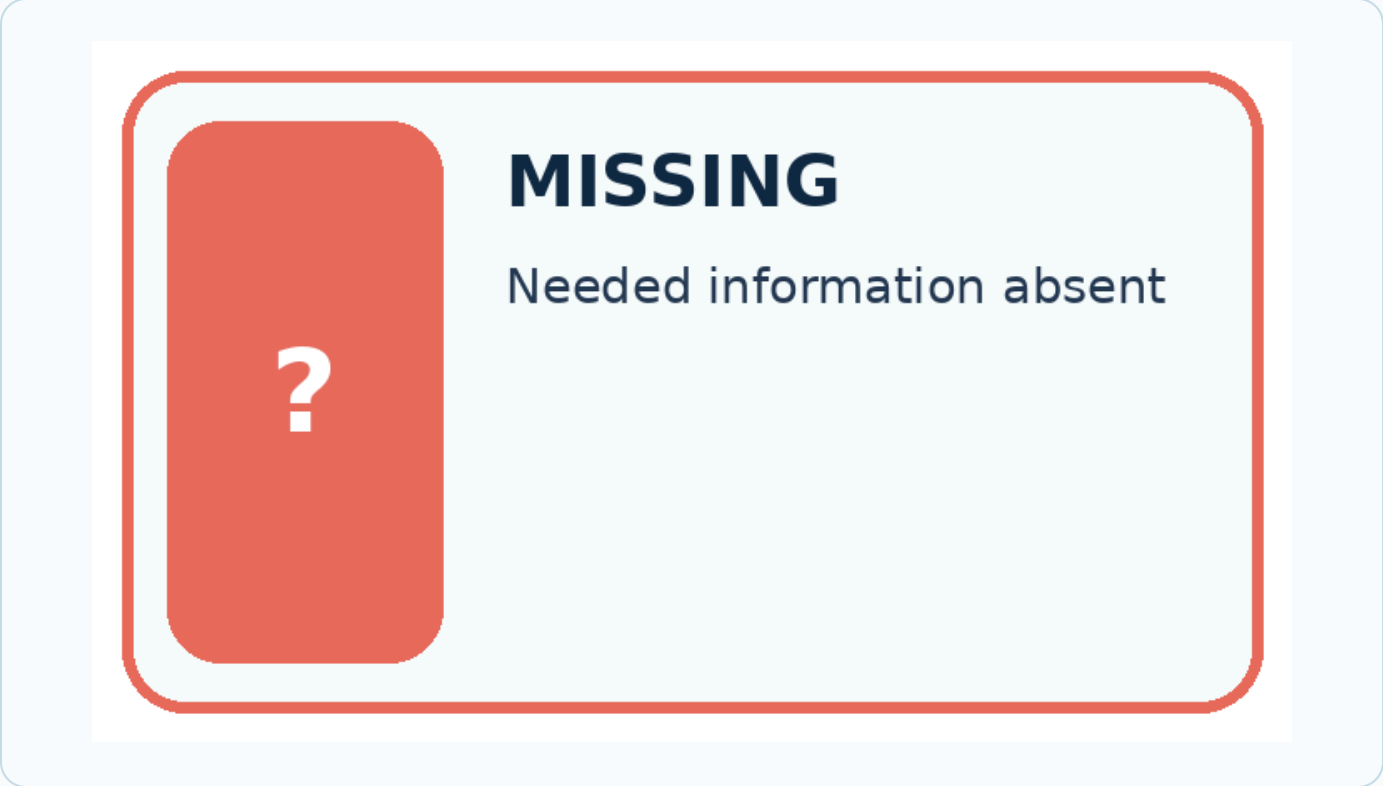
Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.





NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Explain unresolved evidence before the confirmed shelf correction

> A missing recording and a declared-authority record support different conclusions.

The unresolved fifteen/fifty case

Audio transcript reads fifteen boxes; a clear written inventory says fifty boxes. The original audio is unavailable and the assistant reports fifteen as certain.

There is an unresolved quantity conflict and no audio replay. Preserve both sources and ask the responsible person; certainty is unsupported.

Separate the two evidence contracts

Keep two evidence situations separate. The fifteen/fifty-box scenario has no audio replay and no declared authoritative source, so the quantity conflict remains unresolved. The separate shelf scenario supplies a known audio phrase, replay availability and a current inventory declared authoritative for the lookup. Those extra facts support a narrower, auditable correction; they do not prove inventories or transcriptions are universally correct.

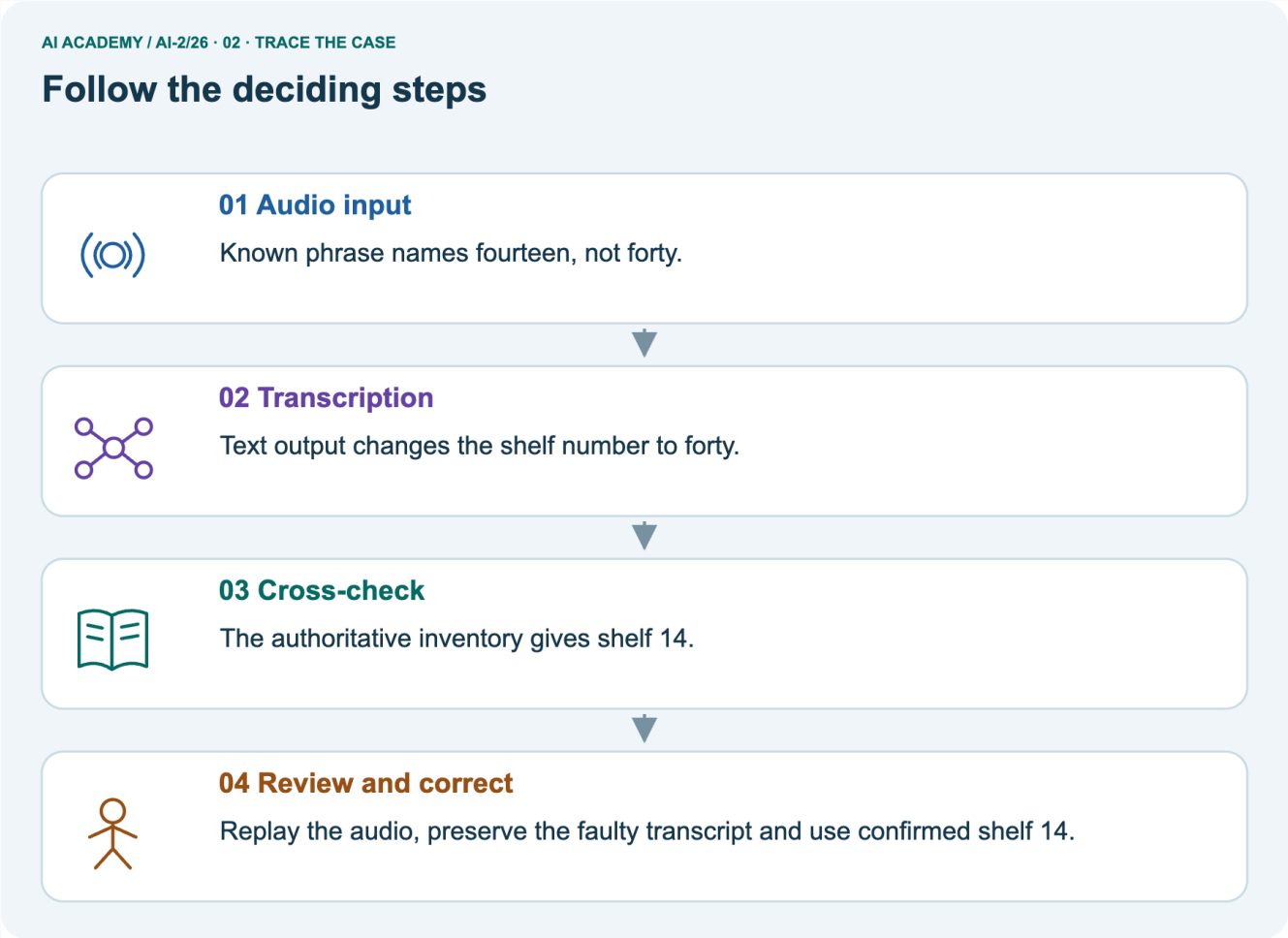
Read the supplied shelf phrase and authority

A fictional librarian dictates a shelf note that passes through **transcription** and catalogue lookup.

Teacher-supplied ground truth audio phrase is “shelf fourteen”. Transcript says “shelf forty”.

A current signed inventory is declared authoritative and lists the requested book on shelf 14. The original audio recording remains available.

Trace first mismatch, cross-check and correction



Follow the labelled picture with the matching explanation. Open the vector diagram

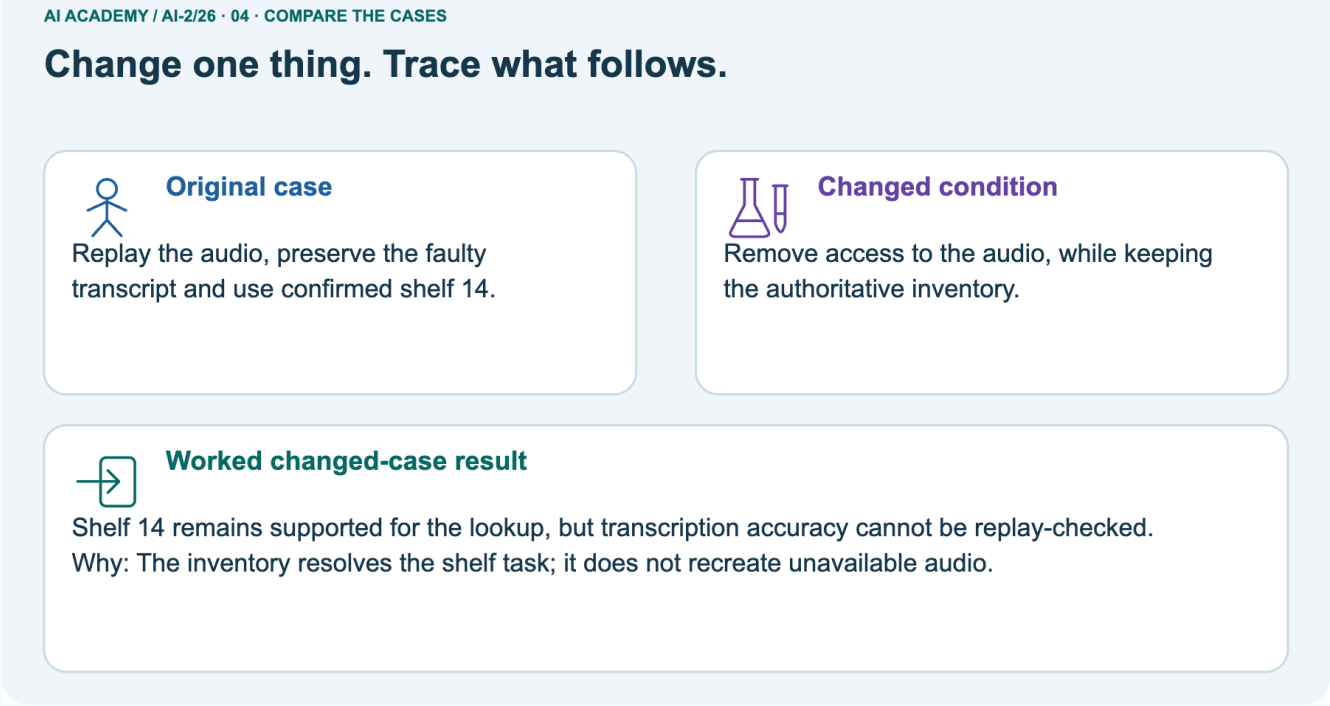
A correct later lookup cannot repair an earlier wrong **input** by magic. Locate the first difference between known audio and transcript, then compare the stated source authority.

Keeping both records makes the correction auditable. A confident spoken answer for shelf 40 would remain wrong despite fluent synthesis.

Predict removal of audio replay only

- Change one thing: Remove access to the audio, while keeping the authoritative inventory. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain provenance and authority limits

- Read the labelled picture: Which stage first fails, and why should the faulty transcript not be silently overwritten?
- Declared authority resolves this task's source choice; it does not make every inventory entry universally accurate.

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

WEEK 26 • WORKED EVIDENCE

1. Voice: sound → sampled signal → recognized words → intent → action.
2. Image: pixels → features → score → label.
3. Noisy sound can produce wrong words even if the intent stage works.
4. Diagnose the first wrong stage before changing later stages.

1. Voice: sound → sampled **signal** → recognized words → intent → action.

2. Image: pixels → features → score → label.

3. Noisy sound can produce wrong words even if the intent stage works.

4. Diagnose the first wrong stage before changing later stages.

Recalculate, retrace, or restate the reasoning in your own representation:

> **Write or draw your reasoning and evidence.**

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

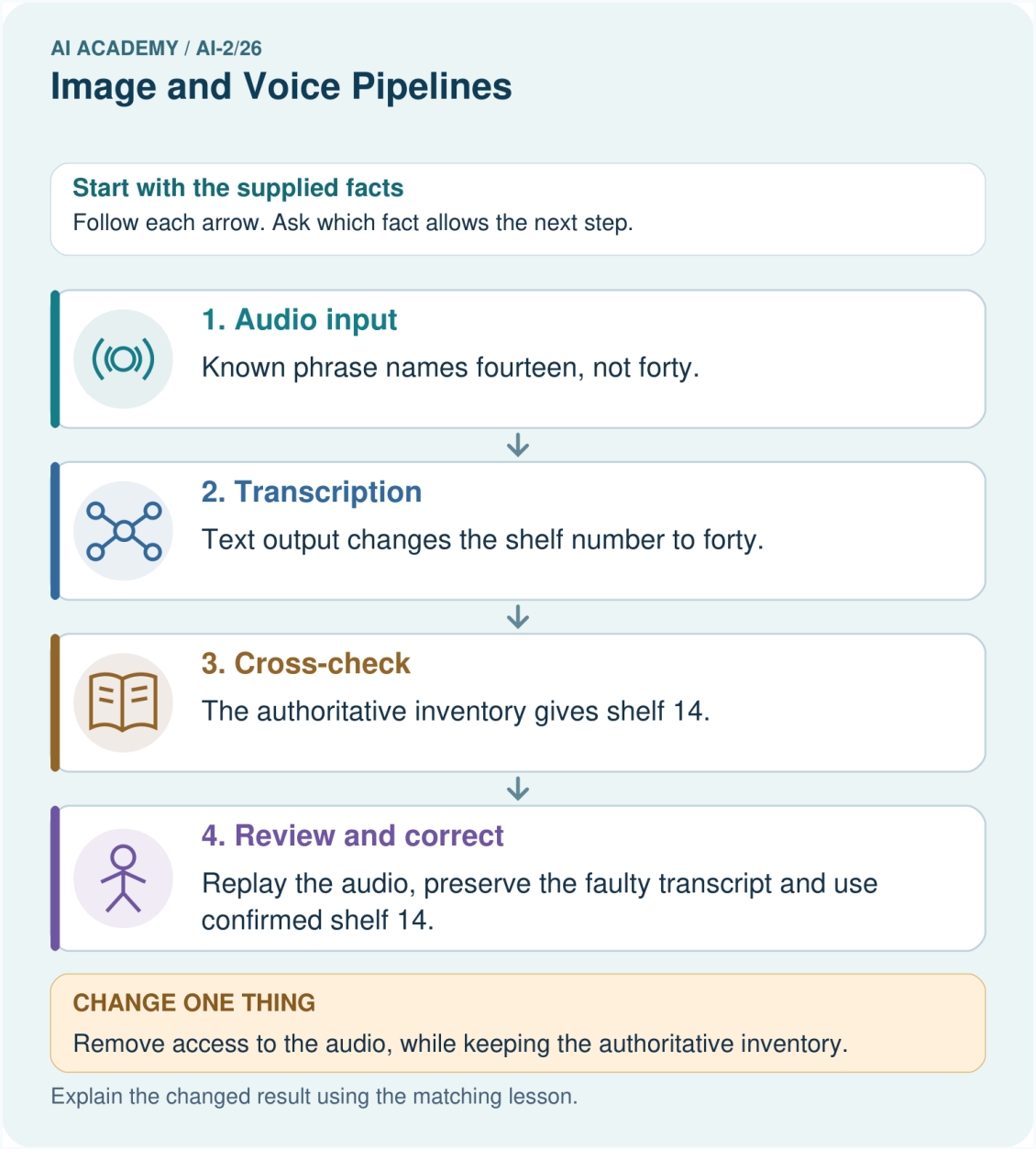
The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Image and Voice Pipelines

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Audio input	Known phrase names fourteen, not forty.
2. Transcription	Text output changes the shelf number to forty.
3. Cross-check	The authoritative inventory gives shelf 14.
4. Review and correct	Replay the audio, preserve the faulty transcript and use confirmed shelf 14.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Design a follow-up that could change the conclusion

> Keep source availability and uncertainty explicit.

Retain support and fresh-case evidence

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

State a revealing changed condition

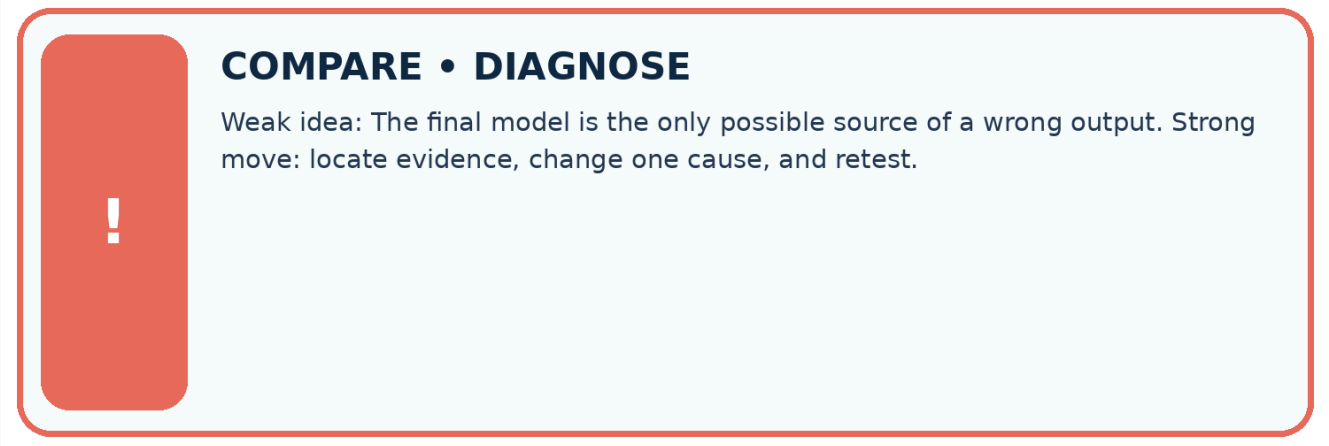
> **Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

▼ **Earlier wording, alternatives and picture transcripts**

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.



Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

> **Write or draw your reasoning and evidence.**

A closer explanation

MAIN LESSON · SECTION 8

Separate a faulty stage from an incomplete decision policy

> A successful later stage cannot recreate missing source evidence.

State the authority and downstream-claim limit

A stage-level accuracy or authority policy does not establish every **modality** agrees or every downstream claim is supported.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Restate the bounded conclusion

> **Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

Trace the standalone W2 conflict

Use workbook W2 for the connected example “Speech **pipeline**s and multimodal **evidence**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Challenge the every-future-case claim

- Use the worked case for Speech **pipeline**s and multimodal **evidence**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the shelf mismatch and retained transcript

Which stage first fails, and why should the faulty transcript not be silently overwritten?

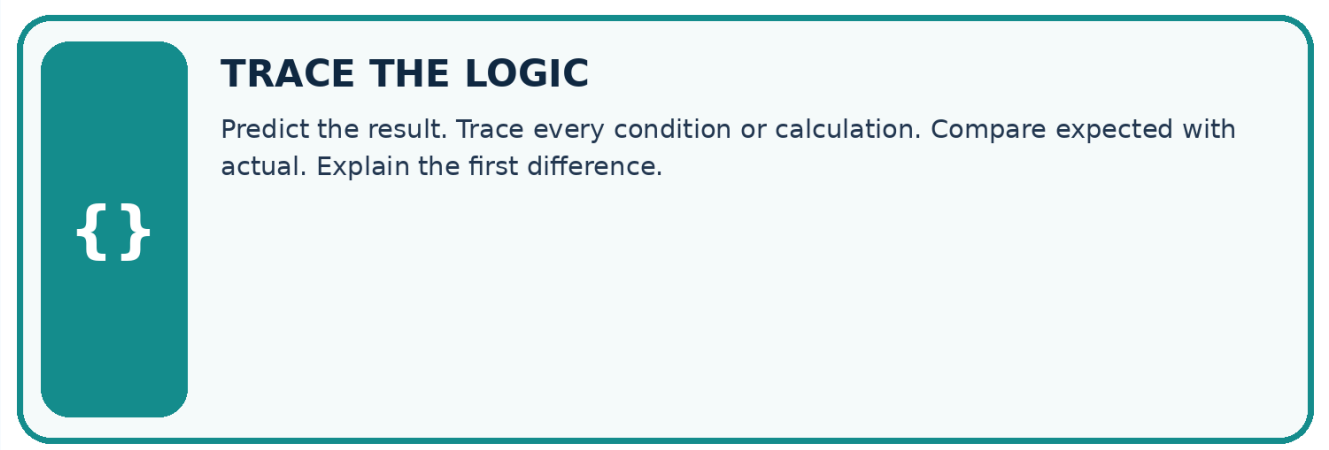
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



SET **evidence**_record PREDICT expected_result RUN one normal case and one boundary case COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST EXPLAIN result, limitation, and next safe step

Trace one **input**. Which line changes the state or chooses a path?

> **Write or draw your reasoning and evidence.**

A closer explanation

Practise with less help: Trace the worked case

Check your reasoning

Explain the picture

> **My explanation using a pictured step and the given facts**

MAIN LESSON · SECTION 9

Audit four complete pipeline records

> A downstream confident answer is not a substitute for source evidence.

The four supplied stage traces

Case	Supplied source and intermediate record	Later output
A · Audio mismatch	Teacher reference phrase shelf fourteen; transcript shelf forty; replay available; authoritative inventory14	Lookup follows transcript and reports40
B · Image mismatch	Shown shelf-label image14; simulated OCR40; no ambiguity in the supplied image	Lookup follows extracted40
C · Unresolved conflict	Transcript15 boxes; written inventory50; original audio unavailable; no authority declared	Assistant reports15 as certain
D · Output mismatch	Known phrase14; transcript14; authoritative inventory14; lookup14	Final spoken/rendered answer says40

Locate, test, preserve and bound the conclusion

1. For each complete case predict the first supported mismatch, or state that the available evidence cannot locate it. Trace source→extraction→interpretation/lookup→final answer; retain each record. Propose one stage-specific test and its expected result, naming which source or replay is available. For caseC do not invent an original recording or assume the inventory is authoritative. For A/B/D distinguish a confirmed mismatched field from its possible technical cause, which the table does not establish. Label work a paper trace; no recogniser, synthesiser or live service is run. Then change one supplied condition, such as replay access, while preserving the rest and recording the new evidence limit. Reserve the roomE/F authority case for independent W4.

Keep observed facts separate from proposed tests

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____

> **Change or keep?:** _____

> **Test / input:** fresh transfer
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

> **Write or draw your reasoning and evidence.**

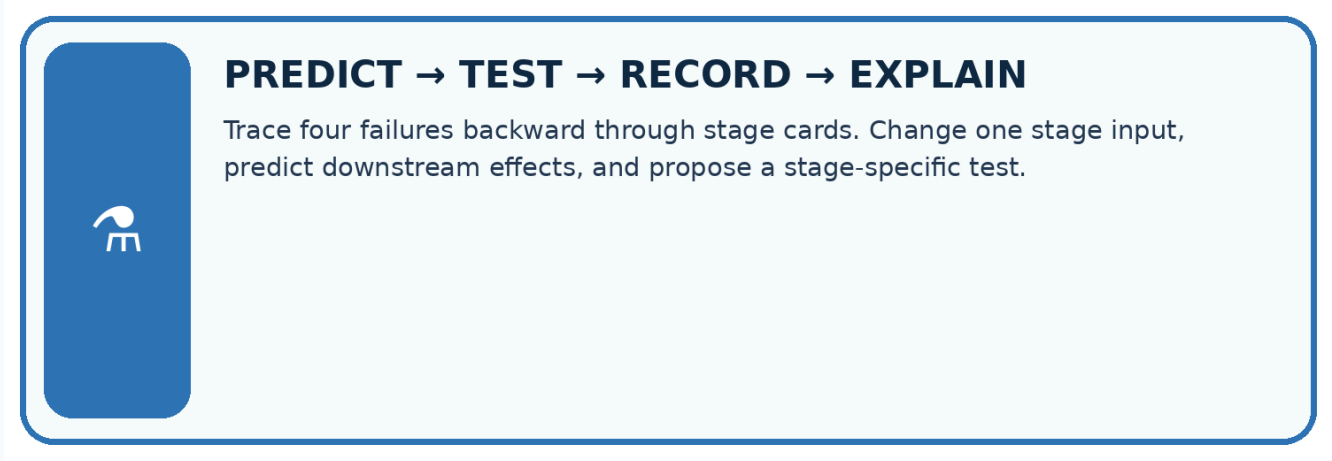
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Use the four supplied pipeline cases. Locate the first supported mismatch or retain an unresolved conflict, propose a stage-specific test and record source/replay/authority limits. These are paper traces, not measured live-service outcomes.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: resolve the approved-room task and state its limit

> Apply the declared authority without claiming perfect transcription.

The independent evidence and task

Transcript says room E; a current written approval says room F. The task requires current approved room. Written approval is declared authoritative; trace the response and the **transcription** limit.

The source and scope reminder

Use only the transcript and current written approval supplied in this new case. The task asks for the approved room and declares which record governs that task; it does not supply verified original speech. Separate the bounded answer from claims about what was actually said.

Your first attempt and support record

> **My independent explanation, trace and limits**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback after the attempt

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Independent case: Apply a new case independently

Week 26 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A voice helper hears find a book, converts speech to text, then searches. It writes find a cook. Which stage first failed and what should you inspect?

My answer, with a trace, calculation or supporting **evidence**
</details>

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

If voice or image **input** is added, show each transformation and a safe fallback.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

A voice request says 'under 90 pages'. The transcript says 'under 19 pages'; filtering correctly finds no books. Identify the first failing stage and a useful confirmation step.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Expose each transformation before using a recognised preference

> Keep the source and evidence boundary beside a recommendation.

This week’s concrete contribution

If the fictional recommender accepts voice or images, show the original supplied input reference, extracted genre/page constraints, lookup criteria and final claim as separate fields. Confirm an important uncertain value before making a hard-constraint claim; preserve the faulty trace and correction. Do not collect personal recordings for this classroom exercise, and do not treat source authority for one task as proof of everything spoken or photographed.

For voice-driven recommendations, confirm important recognised fields before searching or acting and retain source provenance.

Keep the checkpoint and human choices

THIS WEEK’S CHECKPOINT If voice or image **input** is added, show each transformation and a safe fallback.

SYSTEM MAP Reader need → allowed **input**s → transparent rule or score → ranked candidates → explanation → test record → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

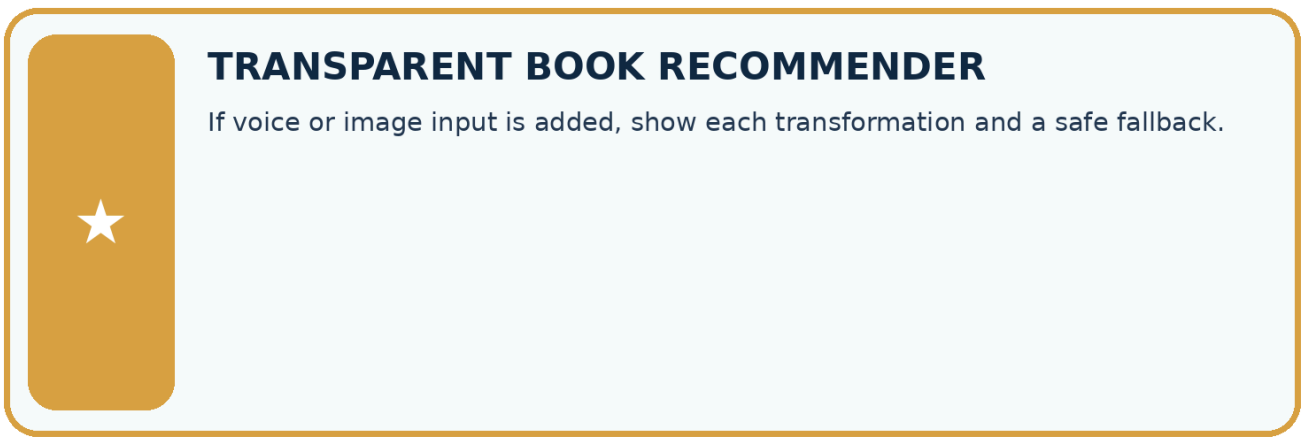
RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits.
Never confuse a ranking with a fact about a person.

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



Evidence for your project

MAIN LESSON · SECTION 16

Finish: justify the stage diagnosis and evidence limit

> Name the available source before claiming what went wrong.

Explain without notes

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of speech **pipeline**s and multimodal **evidence**

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	My own example	Not this / boundary
signal	A measurable quantity carrying information, such as changing sound pressure or light intensity.	
pipeline	A connected sequence of stages in which one stage's output becomes another's input.	
recognition	Interpreting an input as a known category or pattern rather than merely recording its signal.	
stage error	A failure introduced at a particular step of a pipeline.	

Terms for the connected explanation

Word	Meaning
Modality	A form of information such as audio, image or text

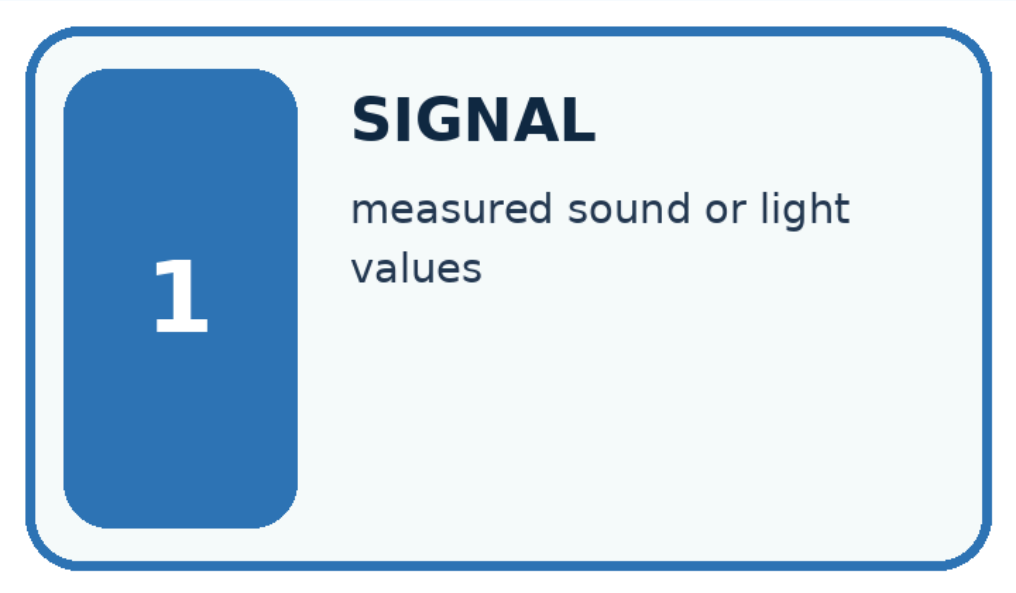
Word	Meaning
Transcription	A text representation of speech
Error propagation	An earlier error affecting later processing

▼ Earlier wording, alternatives and picture transcripts

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.



2

PIPELINE

ordered stages transforming information

3

RECOGNITION

convert signal into a symbol or label



Words used in the closer explanation

▼ **Choose a stronger thinking challenge**

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

- My strongest **evidence** of independent understanding:

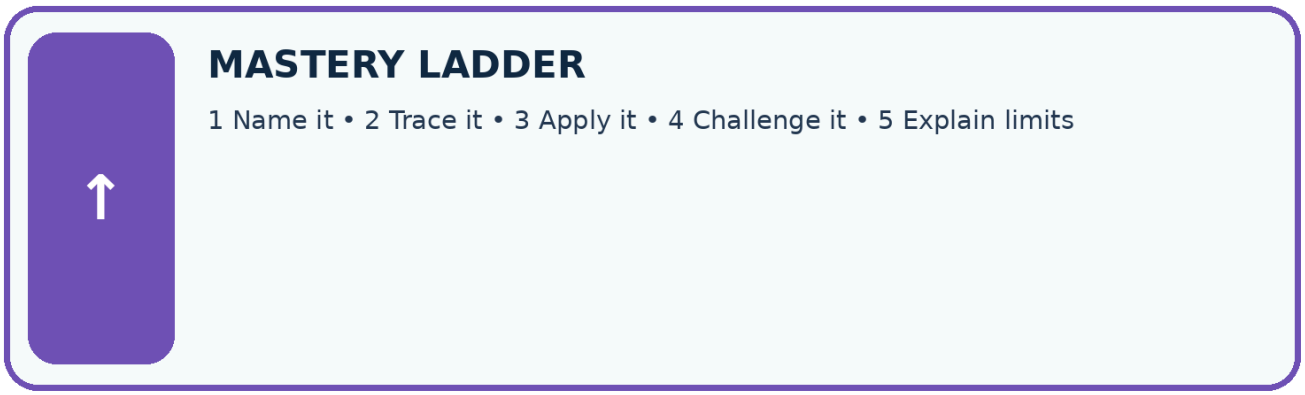
Write or draw your reasoning and evidence.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



▼ Reference: check your explanation

Reference: check your explanation

> Use after practice. Apply a new case independently is the selected independent case.

Check the evidence and its limit

- 1. My fresh example, trace, calculation, or design:
- 2. My **evidence** that it works (include a boundary or missing-data case):
- 3. One limitation, fairness concern, or situation needing human review:

> **Write or draw your reasoning and evidence.**

> **Write or draw your reasoning and evidence.**

Self-check and record

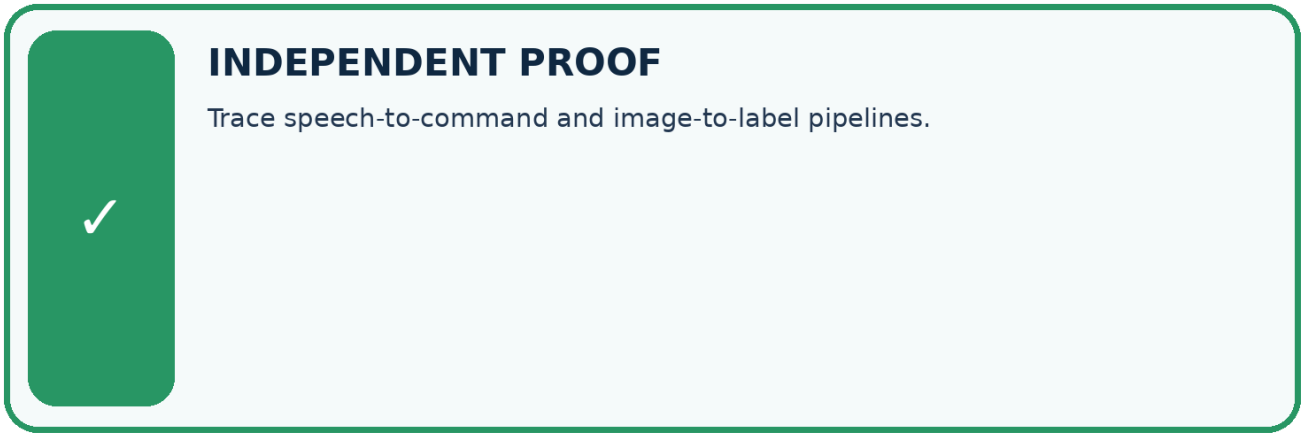
SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

▼ **Earlier wording, alternatives and picture transcripts**

INDEPENDENT MASTERY PROOF

Checklist for explaining this week's idea

Use after practice. Apply a new case independently is the selected independent case.



EXPLANATION CHECK Preserve source, intermediate representations, declared authority and final-claim scope; locate only supported stage failures.

Write or draw your reasoning and evidence.

▼ **Extra practice: Locate the first error in a voice pipeline**

REFERENCE · SECTION 13

Extra practice: Locate the first error in a voice pipeline

> Try the question before opening its worked explanation.

Read the supplied case and question

A voice book helper transcribes speech, extracts preference, then ranks books.

Where can a wrong suggestion begin?

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

Transcription can mishear poetry and pass the wrong preference downstream. Check intermediate results.

Evidence or stage	Value or result
Speech	Reader says poetry
Transcript	May mishear the preference
Extract	Uses the transcript's genre
Rank	Wrong input can produce a wrong suggestion

▼ Earlier wording, alternatives and picture transcripts

Week 26 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Locate the first error in a voice pipeline

A voice book helper transcribes speech, extracts preference, then ranks books.

Where can a wrong suggestion begin?

Transcription can mishear poetry and pass the wrong preference downstream. Check intermediate results.

▼ Reference: original lesson checklist

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Explain the mechanism and terminology of speech pipelines and multimodal evidence	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: A fictional librarian dictates a shelf note that passes through **transcription** and catalogue lookup.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional librarian dictates a shelf note that passes through **transcription** and catalogue lookup.
- Which labelled step matches this detail? Replay the audio, preserve the faulty transcript and use confirmed shelf 14.
- 1. Audio **input**
- 2. Review and correct
- 3. **Transcription**
- 4. Cross-check
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Review and correct Replay the audio, preserve the faulty transcript and use confirmed shelf 14. A correct later lookup cannot repair an earlier wrong **input** by magic.

Locate the first difference between known audio and transcript, then compare the stated source authority. Keeping both records makes the correction auditable.

A confident spoken answer for shelf 40 would remain wrong despite fluent synthesis.

Changed-case question

- Changed-case practice: Remove access to the audio, while keeping the authoritative inventory. Which conclusion is supported by the supplied facts?
- 1. Shelf 14 remains supported for the lookup, but **transcription** accuracy cannot be replay-checked.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

▼ **Check the changed case**

After your attempt, compare with the supported explanation: Shelf 14 remains supported for the lookup, but **transcription** accuracy cannot be replay-checked.

The inventory resolves the shelf task; it does not recreate unavailable audio.

Repair and save your work

Review Text as Pieces (ai-2/25). Goal: Encode and decode text under an explicit toy vocabulary. Short practice: Why is the unknown ! at position 10 rather than position 9?. Return to this week and explain what you changed.

Review Recommendation Scores (ai-2/23). Goal: Compute a weighted score. Short practice: Why does B not become the winner when its score is largest?. Return to this week and explain what you changed.

Review Repair Bias in Data (ai-2/20). Goal: Identify selection and coverage bias. Short practice: Why must 35% be labelled an illustration rather than the measured grade-wide Yes rate?.

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

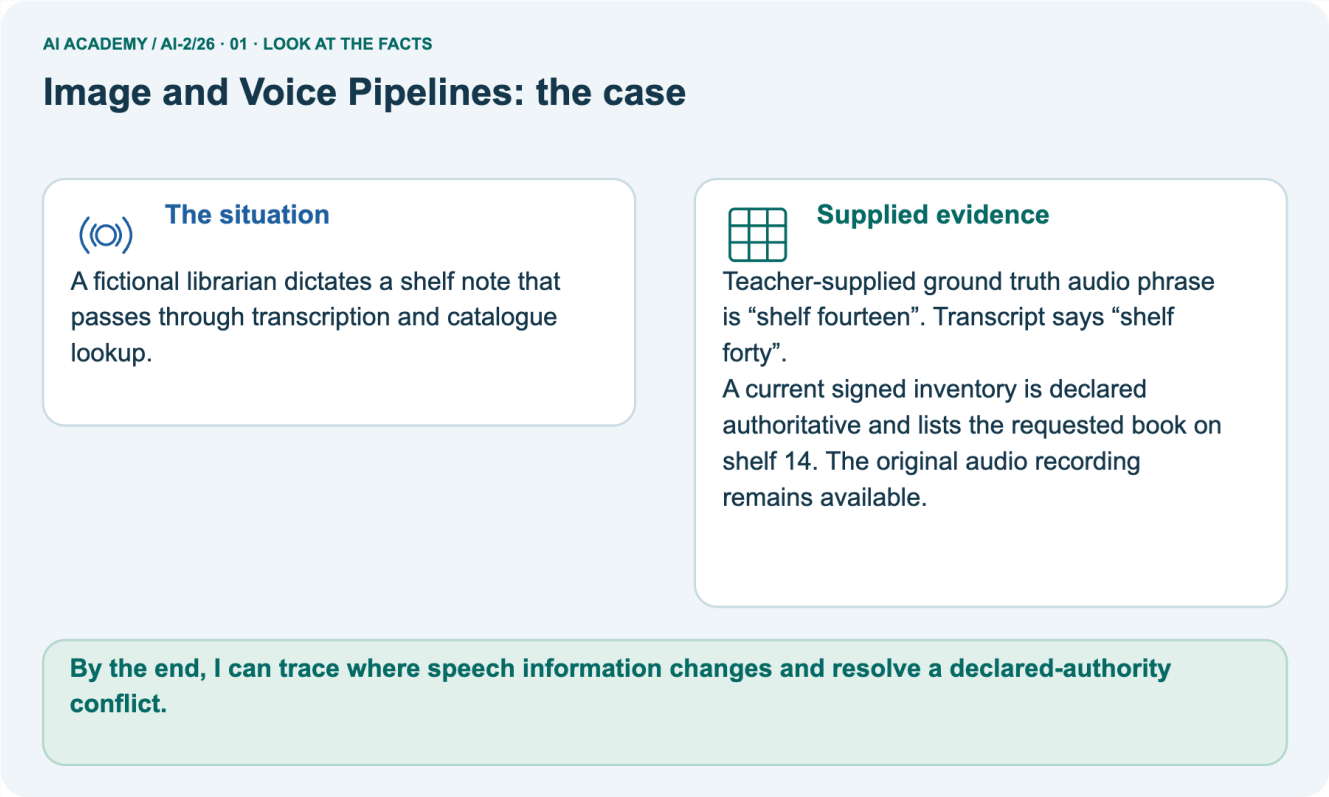
▼ **Reference: the six-picture journey**

REFERENCE · SECTION 18

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Image and Voice Pipelines: the case



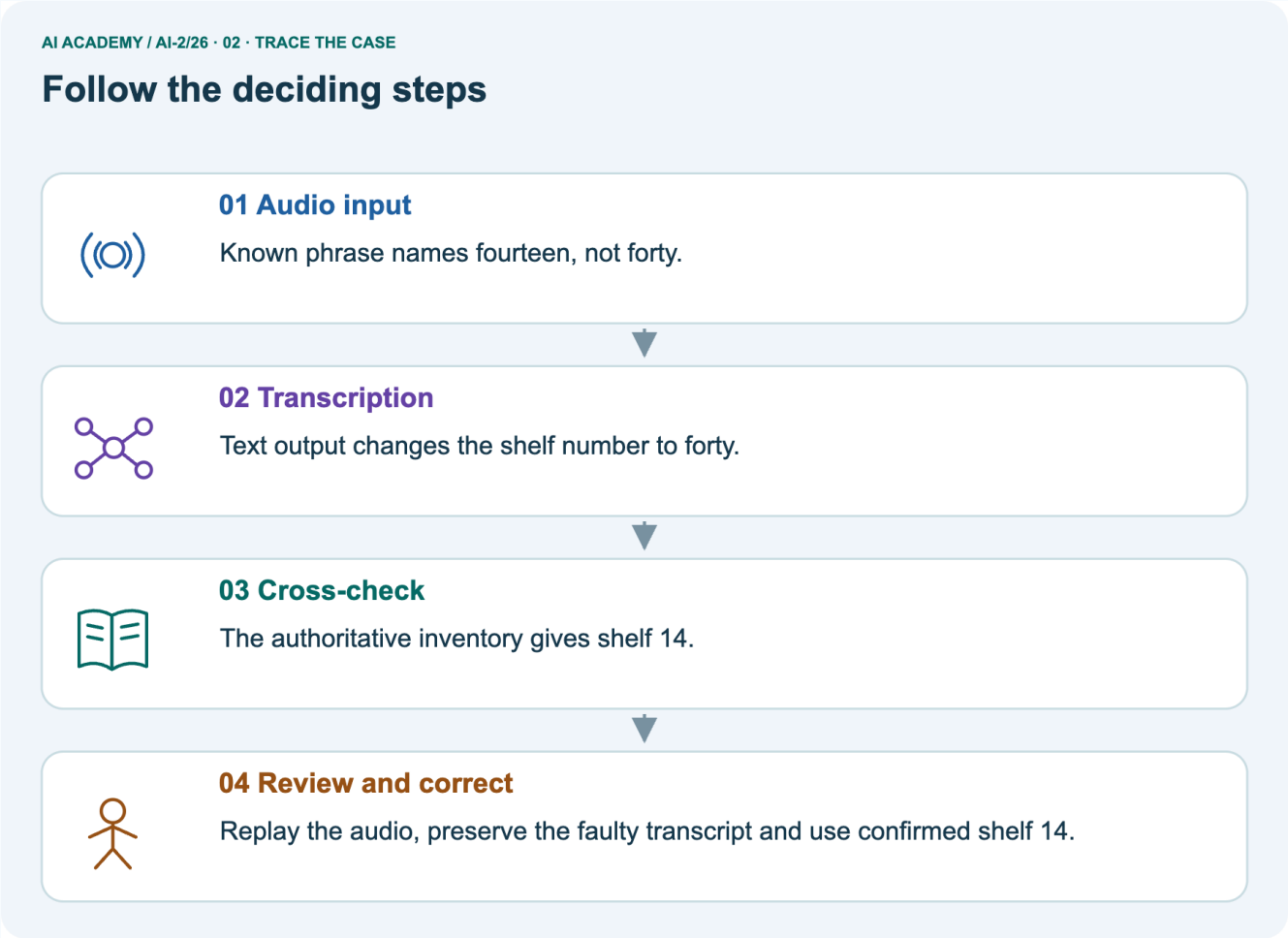
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional librarian dictates a shelf note that passes through **transcription** and catalogue lookup. Supplied facts: Teacher-supplied ground truth audio phrase is “shelf fourteen”. Transcript says “shelf forty”.

A current signed inventory is declared authoritative and lists the requested book on shelf 14. The original audio recording remains available.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

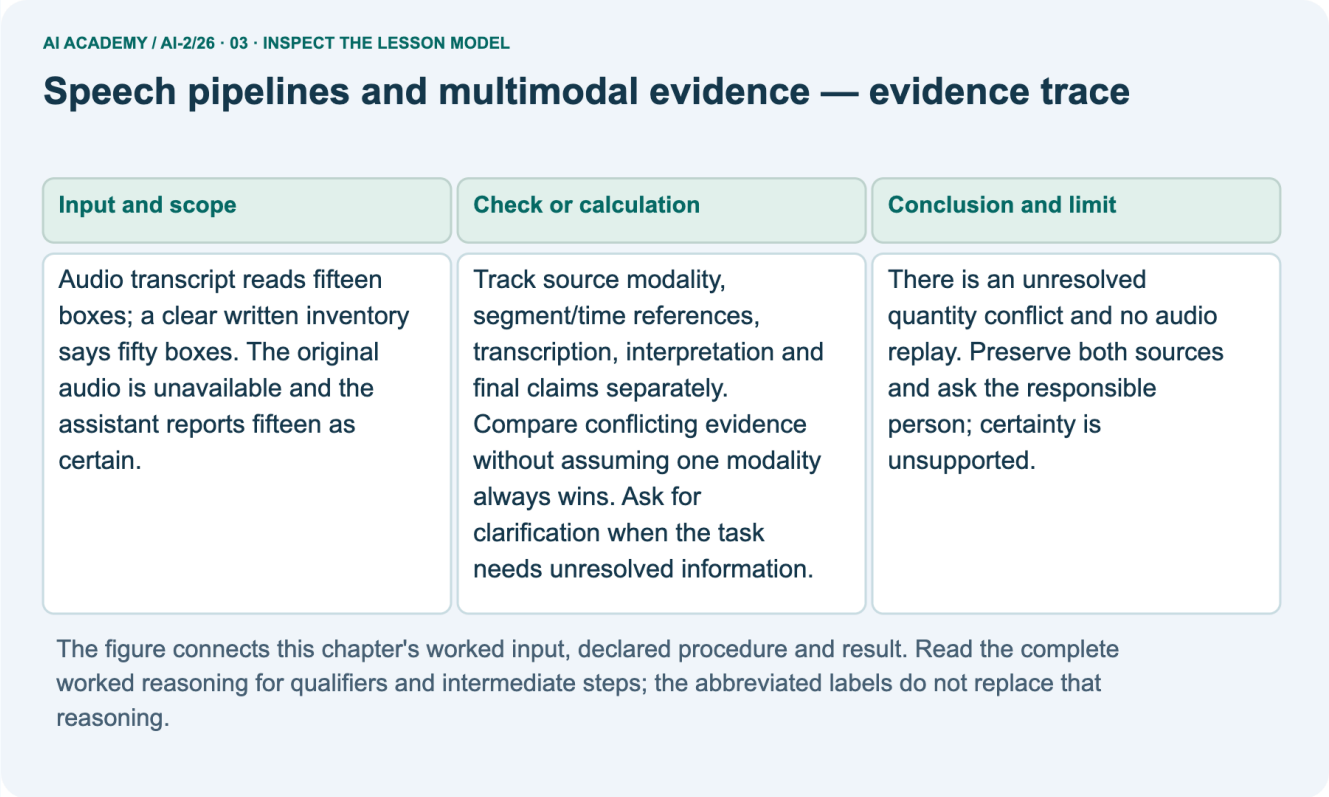


Follow the labelled picture with the matching explanation. Open the vector diagram

Audio **input**: Known phrase names fourteen, not forty. → **Transcription**: Text **output** changes the shelf number to forty. → Cross-check: The authoritative inventory gives shelf 14. → Review and correct: Replay the audio, preserve the faulty transcript and use confirmed shelf 14.

Point to each numbered stage and explain which recorded **input** it uses.

03. Speech pipelines and multimodal evidence — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Speech **pipeline**s and multimodal **evidence** — **evidence** trace. Audio transcript reads fifteen boxes; a clear written inventory says fifty boxes.

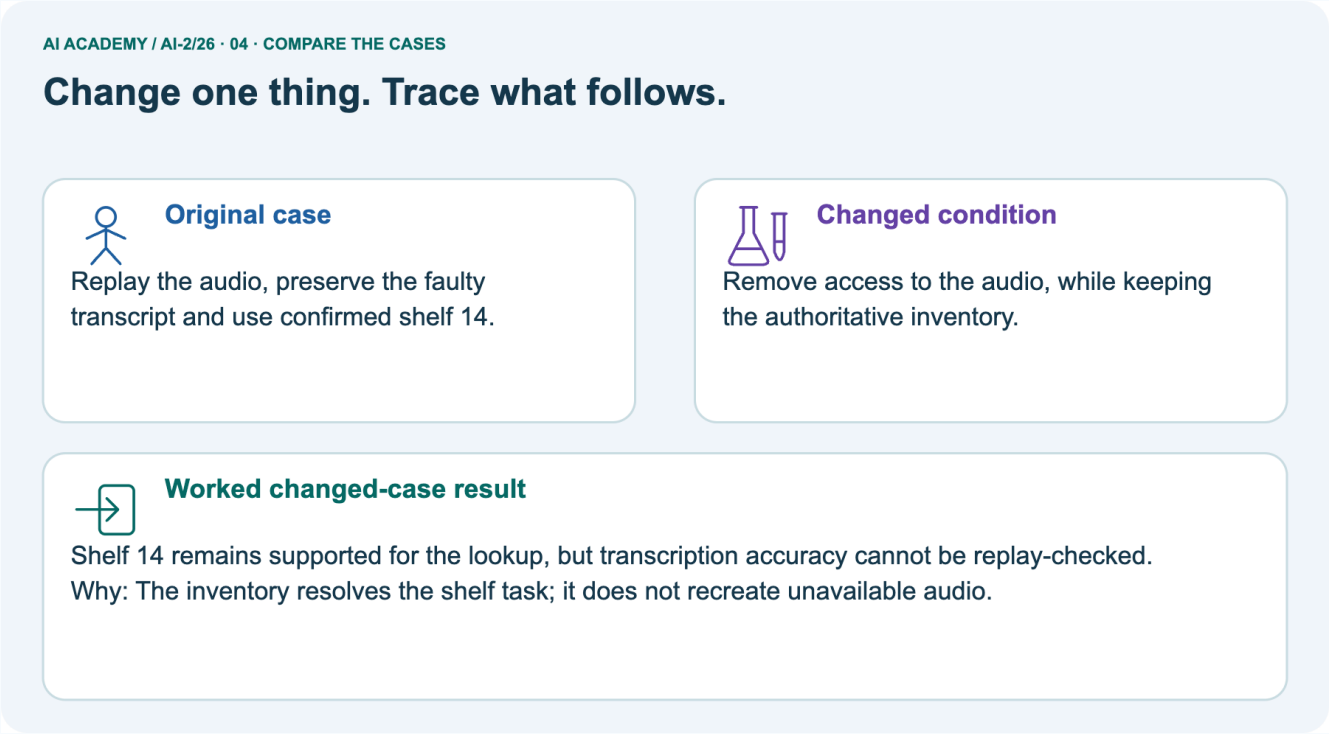
The original audio is unavailable and the assistant reports fifteen as certain. | Track source **modality**, segment/time references, **transcription**, interpretation and final claims separately.

Compare conflicting **evidence** without assuming one **modality** always wins. Ask for clarification when the task needs unresolved information. | There is an unresolved quantity conflict and no audio replay.

Preserve both sources and ask the responsible person; certainty is unsupported.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



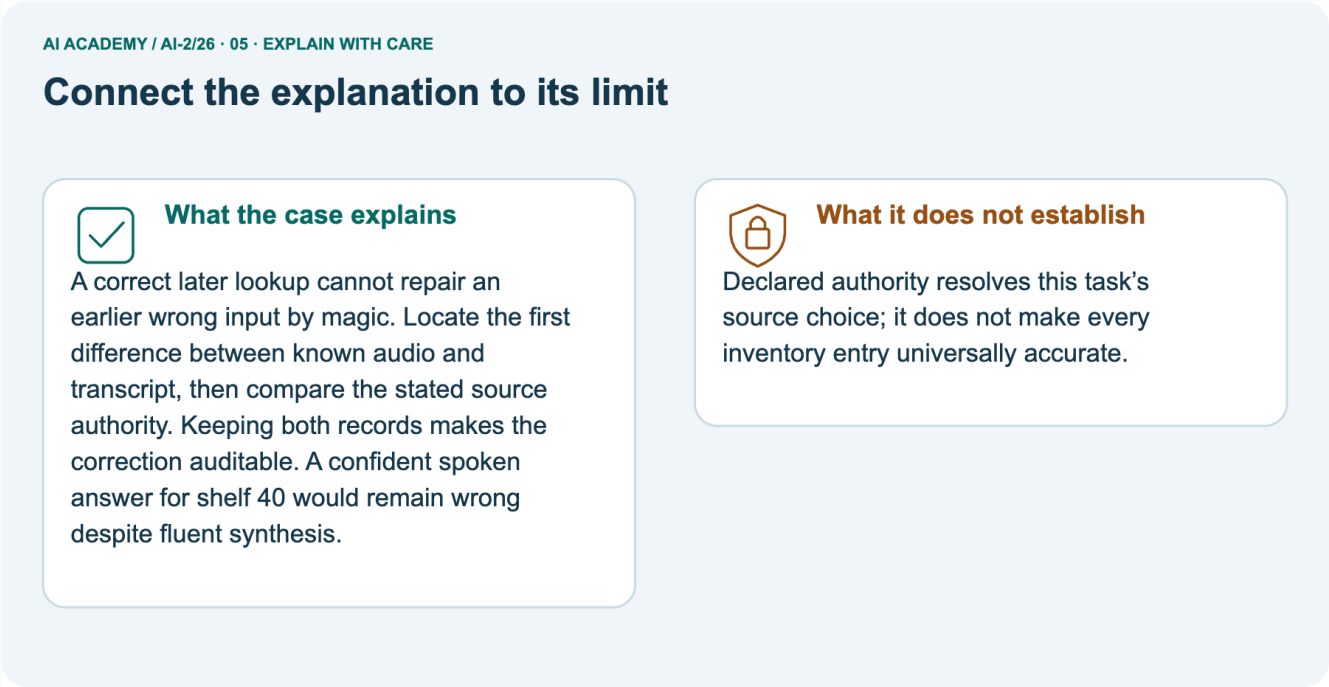
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Replay the audio, preserve the faulty transcript and use confirmed shelf 14.
Changed condition: Remove access to the audio, while keeping the authoritative inventory.

Expected result: Shelf 14 remains supported for the lookup, but **transcription** accuracy cannot be replay-checked. Reason: The inventory resolves the shelf task; it does not recreate unavailable audio.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

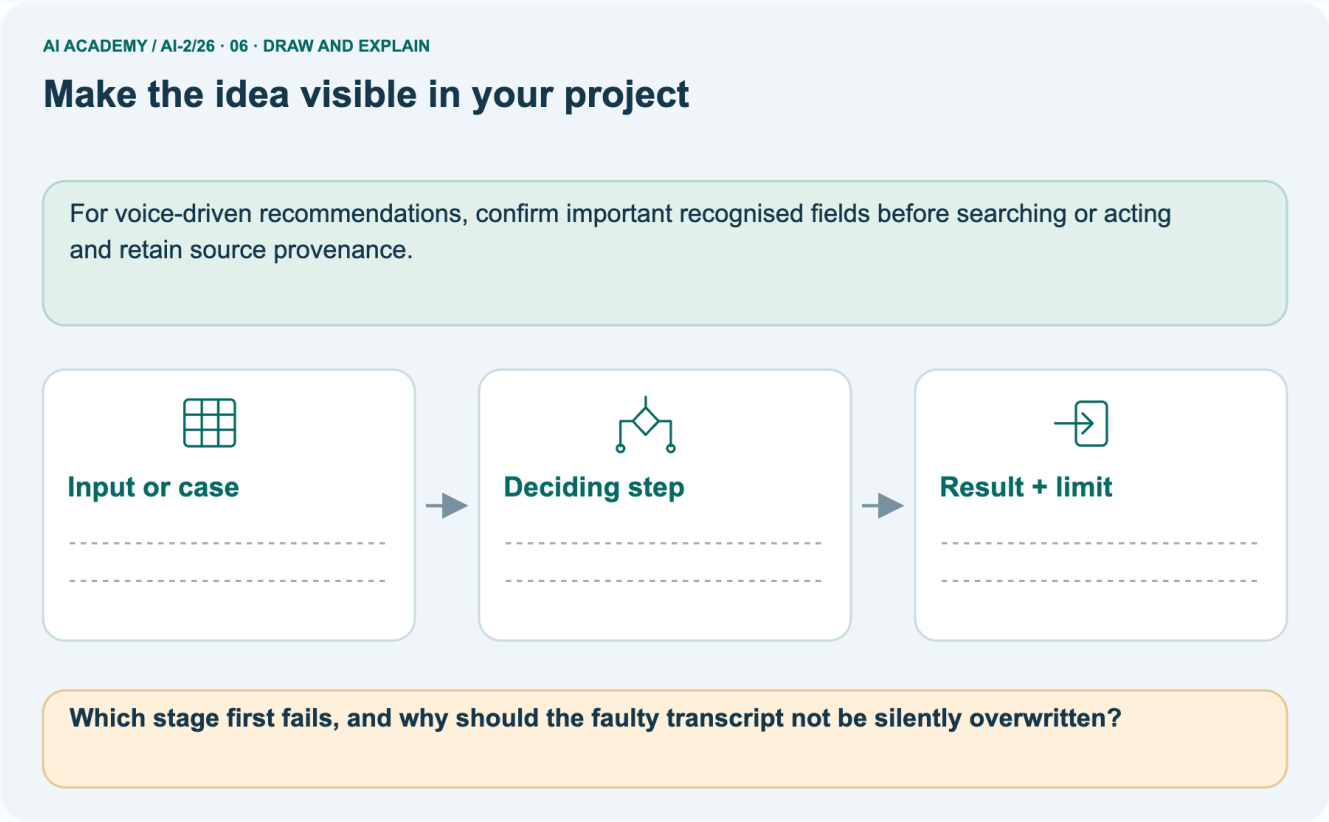
A correct later lookup cannot repair an earlier wrong **input** by magic. Locate the first difference between known audio and transcript, then compare the stated source authority.

Keeping both records makes the correction auditable. A confident spoken answer for shelf 40 would remain wrong despite fluent synthesis.

Limit: Declared authority resolves this task's source choice; it does not make every inventory entry universally accurate.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For voice-driven recommendations, confirm important recognised fields before searching or acting and retain source provenance. Draw your **input**, deciding step, result and limit.

Supported question: Which stage first fails, and why should the faulty transcript not be silently overwritten?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Image and Voice Pipelines: the case

02. Follow the deciding steps

03. Speech pipelines and multimodal evidence — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.